

PHISHING UNFOLDING

SOC Alert Triage & Investigation Report

TryHackMe SOC Simulator • Simulated Security Operations Environment

Executive Summary. This case study documents the Phishing Unfolding scenario completed in the TryHackMe SOC Simulator. The exercise places the analyst in a simulated corporate environment where a phishing attack unfolds over time. The stated task is to monitor and analyze alerts, identify critical events such as PowerShell execution, reverse shell connections, and suspicious DNS requests, and prepare case documentation describing the observed activity.

1. Scenario & Objectives

The scenario is designed as a 40-minute, medium-difficulty SOC simulation. The analyst's responsibility is to follow the attack as it unfolds, correlate security alerts, identify significant events, and document the activity in a case report.

Key objectives

- Monitor and analyze real-time security alerts as the simulated phishing attack progresses.
- Identify and document critical events, including PowerShell executions, reverse shell connections, and suspicious DNS requests.
- Build a coherent picture of the activity from individual alerts and record findings for the security team.

2. Investigation Approach

Phase	Activity	Purpose
01	Monitor	Review incoming alerts and establish the sequence of events.
02	Triage	Determine which events require deeper investigation based on the available alert context.
03	Correlate	Connect related endpoint, network, and DNS activity to understand the developing attack chain.
04	Document	Record important observations, evidence, and conclusions in the case report.
05	Assess	Summarize the observed scope and identify the critical events requiring response.

3. Scenario Focus

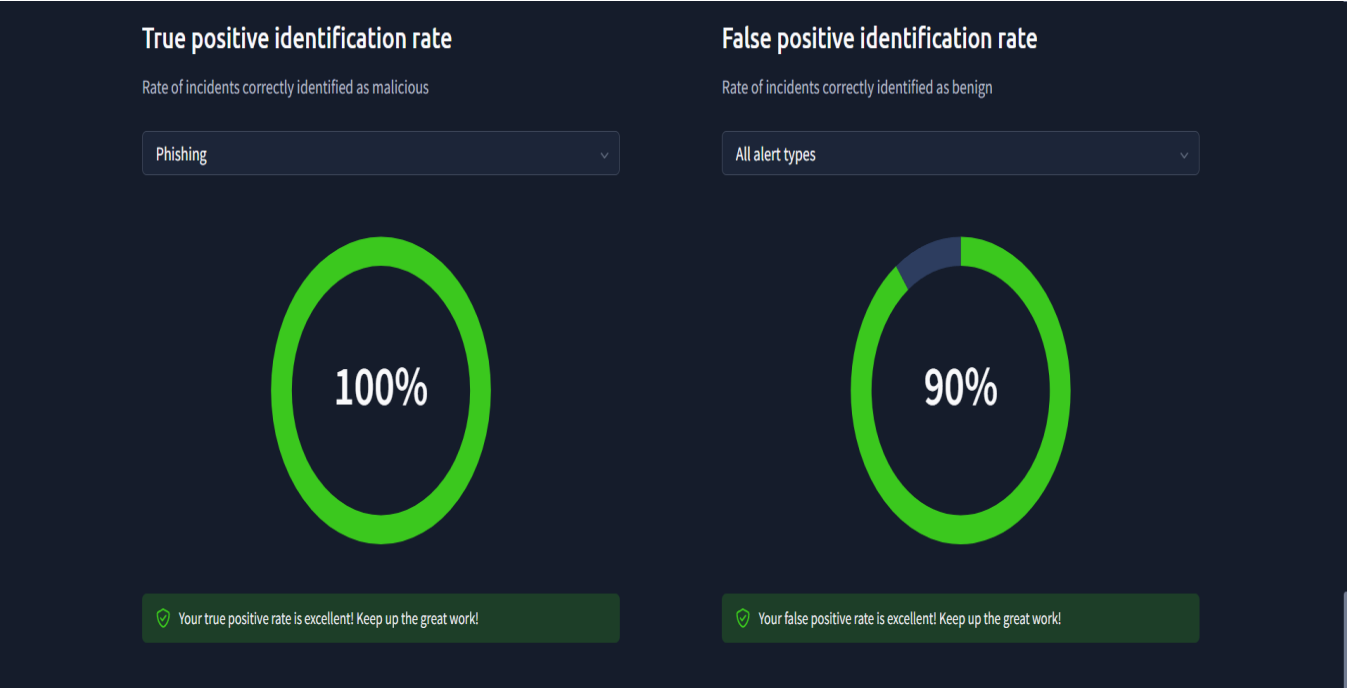
PowerShell Activity	Reverse Shell	Suspicious DNS
Endpoint execution requiring investigation	Network connection requiring investigation	Potentially malicious DNS activity requiring investigation

These three event categories are explicitly identified by the scenario as critical investigation points. The report therefore treats them as investigation objectives rather than assuming details that are not shown in the supplied evidence.

4. Simulator Performance & Evidence

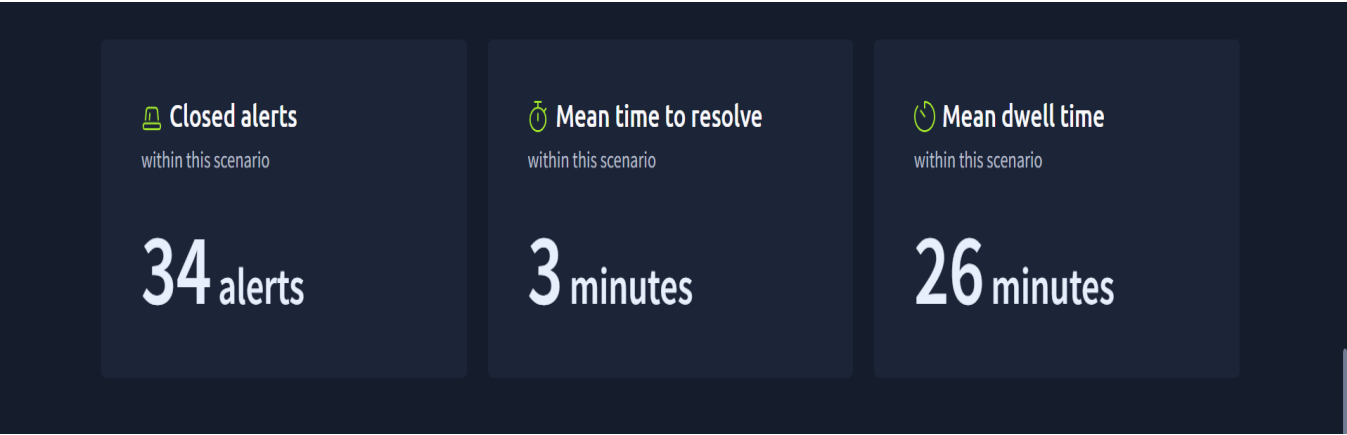
34	3 min	26 min
Closed alerts	Mean time to resolve	Mean dwell time
100%	90%	Phishing
True-positive identification for evaluated phishing scenario	Benign-alert identification across evaluated alert set	Selected alert scenario

Evidence 01 — Identification Metrics



The supplied simulator screenshot shows a 100% true-positive identification rate for the selected phishing scenario and a 90% benign-alert identification value in the dashboard. The interface labels the latter as the rate of incidents correctly identified as benign.

Evidence 02 — Alert Handling Metrics



The second supplied screenshot reports 34 closed alerts, a 3-minute mean time to resolve, and a 26-minute mean dwell time within the scenario.

5. Analyst Takeaways

- Alert triage is not limited to recognizing an alert; the analyst must examine context and determine whether activity is malicious or benign.
- Phishing investigations can develop into endpoint and network investigations, making correlation between different alert types important.
- PowerShell execution, reverse shell connections, and suspicious DNS requests are high-value events that should receive focused investigation in this scenario.
- Documenting the attack progression creates a clear handoff for other analysts and helps the security team understand the broader incident.

6. Skills Demonstrated

SOC Operations: Security alert monitoring, alert triage, malicious/benign classification, incident investigation, case documentation, incident resolution. Investigation Focus: Phishing, PowerShell activity, reverse shell activity, suspicious DNS requests, attack-chain correlation. Reporting: Evidence-based findings, performance metrics, structured incident documentation.

7. Conclusion

The Phishing Unfolding scenario provided practical exposure to an L1-style SOC workflow in a controlled simulation: monitor alerts, triage important events, correlate activity, identify malicious versus benign behavior, and document the resulting case. The simulator also provided measurable performance indicators that can be retained as portfolio evidence alongside the detailed case report.